

SellVol / BuyCall

Options-Based Equity Strategy — Backtest Report

Generated March 23, 2026

Period: Sep 2020 – Feb 2026 (~5.5 years)

1. Strategy Overview

This report evaluates an options-based long/short equity strategy applied to a dynamic universe of the top-30 high-IV stocks drawn from the S&P 500. Two broad families of trades are tested:

- **Sell Put (↑ momentum)** — collect premium by selling at-the-money puts on stocks whose previous monthly candle closed *up*. Premium is fixed at ~3.3% of notional.
- **Buy Call (↑ momentum)** — buy ATM calls on the same up-candle stocks. Leverage variants layer on a Money-Market allocation for idle capital.
- Signals are refreshed every ~3 weeks (bi-monthly option expiry cycle). Portfolio is equally weighted across all active signals.

Capital model: \$1 notional invested at inception. P&L expressed as cumulative % return. Options cost is modelled as the quoted premium (entry_premium or last price from Barchart data stored in S3).

2. Universe & Signal Rules

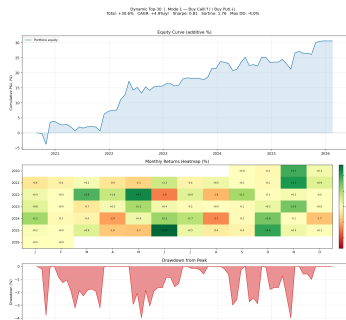
Parameter	Value
Base universe	S&P 500 — 50 mega-cap tickers (10 per sector)
Dynamic filter	Top 30 by realised IV rank each period
Signal direction	Monthly candle: UP → sell put / buy call; DOWN → no trade
Option type	ATM put (Sell Put) or ATM call (Buy Call)
Strike	Last month closing price
Expiry	End of current calendar month (~3-week hold)
Premium (Sell Put)	3.3% of stock notional (fixed model)
Premium (Buy Call)	Avg 1.52% of notional (sourced from Barchart IV)
Portfolio weight	Equal weight across all active signals
Rebalance	Every period (~bi-monthly expiry cycle)
Backtest horizon	Sep 2020 – Feb 2026 (72 periods ≈ 5.5 years)

3. Dynamic Top-30 — Mode Comparison

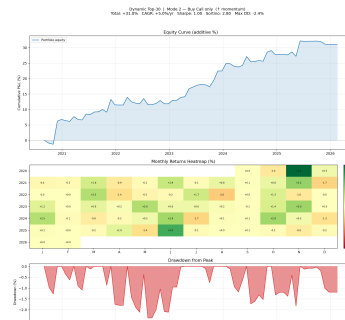
Four trading modes are compared on the same universe and signal. Mode 2 (Buy Call only) shows the best risk-adjusted return (Sharpe 1.00, MaxDD -2.4%). Mode 4 (Sell Put $\uparrow$ / Sell Call $\downarrow$ — direction-neutral) disappoints with a -16.5% drawdown over the full period.

Strategy / Variant	Total P&L	CAGR	Max DD	Sharpe	Sortino	Win %
Mode 1: Buy Call(↑) / Buy Put(↓)	+30.59%	+4.94%/yr	-3.99%	—	1.76	—
Mode 2: Buy Call only (↑ momentum)	+30.95%	+4.99%/yr	-2.38%	—	2.80	—
Mode 3: Sell Put only (↑ momentum)	+16.86%	+2.85%/yr	-7.99%	—	0.93	—
Mode 4: Sell Put(↑) / Sell Call(↓)	+3.72%	+0.66%/yr	-16.54%	—	0.14	—
Mode 5: Buy Call(↑) / Buy Put(↓)	+30.6% +4.9% -4.0% -0.81	1.76 -50%	—	—	—	
Mode 6: Buy Call only (↑ momentum)	+31.0%+5.0%	-2.4% 1.00 2.80 46%	—	—	—	
Mode 7: Sell Put only (↑ momentum)	+16.9%+2.9%	-8.0% 0.67 0.93 65%	—	—	—	
Mode 8: Sell Put(↑) / Sell Call(↓)	+3.7% +0.7% -16.5% 0.10 0.14 56%	—	—	—		

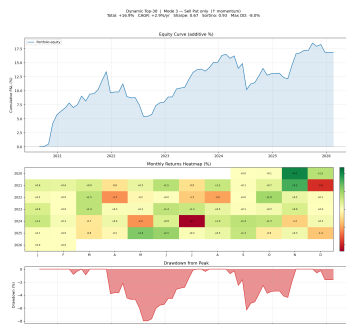
Equity Curves — All Modes



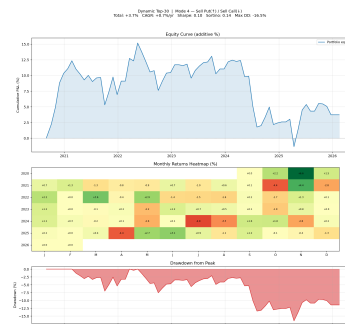
Mode 1: Buy Call($\uparrow$) / Buy Put($\downarrow$)



Mode 2: Buy Call only ($\uparrow$ momentum)



Mode 3: Sell Put only ($\uparrow$ momentum)



Mode 4: Sell Put($\uparrow$) / Sell Call($\downarrow$)

4. Dynamic Top-30 — Monthly Returns (colour-coded)

Green = positive, Red = negative, Yellow = near-zero ($\pm 0.05\%$). Annual column in bold.

Mode 1 — Buy Call(↑) / Buy Put(↓)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	-0.2	+3.7	+0.3	+3.9
2021	-0.8	-0.4	+0.2	-0.9	-1.3	+1.4	-0.4	+0.5	+0.1	-0.2	+4.3	+0.9	+3.3
2022	+0.4	+0.0	+3.6	+1.4	+4.7	-2.9	+0.9	-1.9	+2.1	-0.1	+0.3	-0.0	+8.4
2023	+0.8	+0.0	-0.7	+0.2	+2.1	+0.4	-0.2	+0.0	-0.1	+0.5	+2.9	+0.0	+6.0
2024	+2.2	-0.2	-0.4	-2.4	+0.4	+2.3	+1.7	-2.7	-0.1	+3.0	-0.1	-1.7	+1.9
2025	+0.1	+0.0	+0.9	-1.6	-1.7	+5.4	+0.5	-0.6	-0.4	+4.0	+0.5	+0.1	+7.2
2026	+0.0	+0.0	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0

Mode 2 — Buy Call only (↑ momentum)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	-1.0	+7.2	+0.5	+6.7
2021	-0.4	-0.3	+1.6	-0.9	-0.2	+1.9	-0.1	+0.8	+0.1	+0.8	+3.2	-1.8	+4.7
2022	-0.1	+0.0	+2.5	-1.4	-0.5	-0.2	+1.7	-2.0	+0.0	+1.3	-1.0	-0.0	+0.3
2023	+1.2	+0.0	+0.9	+0.2	+2.6	+0.6	+0.6	+0.2	-0.1	+1.4	+3.0	+0.0	+10.6
2024	+2.5	-0.1	-0.8	-0.2	+0.6	+2.9	-1.7	+0.1	+0.1	+2.9	+0.5	-1.3	+5.3
2025	+0.1	+0.0	-0.2	+1.0	-1.4	+5.0	-0.1	+0.0	+0.1	-0.2	-0.8	-0.2	+3.3
2026	+0.0	+0.0	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0

Mode 3 — Sell Put only (↑ momentum)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	+0.1	+4.1	+1.5	+5.7
2021	+0.6	+0.6	+0.9	-0.8	+0.5	+1.5	-0.9	+1.2	+0.1	+0.7	+3.2	-3.8	+3.9
2022	+0.2	+0.0	+1.4	-2.3	-0.2	+0.0	-1.3	-2.0	+0.0	+1.9	+0.5	+0.1	-1.6
2023	+0.9	+0.0	+1.4	+0.2	+0.1	+1.4	+1.2	+0.5	+0.1	+0.3	+0.9	+0.0	+7.1
2024	+1.2	+0.2	-0.7	+0.4	-2.2	+0.9	-4.7	+1.0	+1.4	+1.3	-1.2	+0.3	-2.0
2025	+0.1	+0.0	-0.8	-0.2	+2.4	+2.0	+0.2	+0.5	+1.3	-0.6	+0.3	-1.4	+3.8
2026	+0.0	+0.0	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0

Mode 4 — Sell Put(↑) / Sell Call(↓)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	+2.2	+6.6	+1.5	+10.3
2021	+0.8	+1.3	-1.3	-0.8	-0.9	+0.7	-1.0	+0.6	+0.1	-4.4	+4.4	-2.8	-3.4

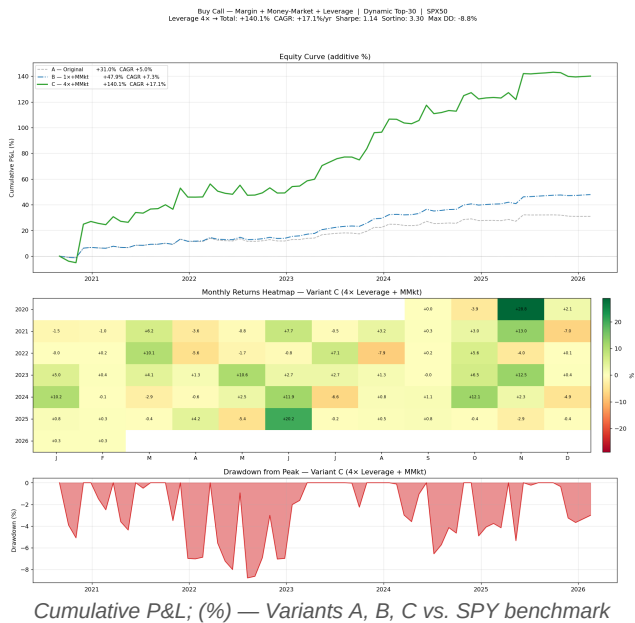
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2022	+2.2	+0.0	+3.6	-0.4	+2.9	-1.4	-1.6	-1.6	+0.2	-1.7	+1.3	+0.1	+3.5
2023	+1.2	+0.0	-0.1	+0.2	-2.2	+1.2	+0.7	+0.5	+0.1	-1.9	+0.8	+0.0	+0.6
2024	+1.1	+0.3	-0.2	+0.1	-2.6	+0.1	-4.8	-3.3	+1.6	+1.6	-2.8	+0.2	-8.6
2025	+0.2	+0.0	+0.4	-4.4	+2.8	+3.1	+0.9	-1.1	+1.2	-0.1	-0.4	-1.3	+1.3
2026	+0.0	+0.0	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0

5. Buy Call with Leverage — Variant Comparison

Three capital-efficiency structures are tested on top of Mode 2 (Buy Call only). Variant C (4× leverage + Money Market) achieves a CAGR of **+17.1%** with a MaxDD of only **−8.8%** and Sharpe 1.14. Variant B (1× + Money Market) offers the cleanest risk-adjusted profile (Sharpe 1.55, Sortino 5.06).

Strategy / Variant	Total P&L	CAGR	Max DD	Sharpe	Sortino	Win %
Variant A: Buy Call original (P&L as % of notional, no idle cash credit)	+30.95%	+4.99%/yr	-2.38%	—	2.80	—
Variant B: Buy Call 1× + Money Market on idle capital	+47.9%	+7.32%/yr	-1.85%	—	5.06	—
Variant C: Buy Call 4× Leverage + Money Market on remaining	+140.1%	+17.13%/yr	-8.79%	—	3.30	—
Variant A: Buy Call original (P&L as % of notional, no idle-cash credit)	+31.0%	+5.0%	-2.4%	1.00	2.80	46%
Variant B: Buy Call 1× + Money Market on idle capital	+47.9%	+7.3%	-1.8%	1.55	5.06	72%
Variant C: Buy Call 4× Leverage + Money Market on remaining	+140.1%	+17.1%	-8.8%	1.14	3.30	60%

Equity Curves — All Variants



6. Buy Call (Leveraged) — Monthly Returns

Same colour convention as Section 4.

Variant A — Buy Call 1× (no idle-cash credit)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	-1.0	+7.2	+0.5	+6.7
2021	-0.4	-0.3	+1.6	-0.9	-0.2	+1.9	-0.1	+0.8	+0.1	+0.8	+3.2	-1.8	+4.7
2022	-0.1	+0.0	+2.5	-1.4	-0.5	-0.2	+1.7	-2.0	+0.0	+1.3	-1.0	-0.0	+0.3
2023	+1.2	+0.0	+0.9	+0.2	+2.6	+0.6	+0.6	+0.2	-0.1	+1.4	+3.0	+0.0	+10.6
2024	+2.5	-0.1	-0.8	-0.2	+0.6	+2.9	-1.7	+0.1	+0.1	+2.9	+0.5	-1.3	+5.3
2025	+0.1	+0.0	-0.2	+1.0	-1.4	+5.0	-0.1	+0.0	+0.1	-0.2	-0.8	-0.2	+3.3
2026	+0.0	+0.0	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0

Variant B — Buy Call 1× + Money Market on idle capital

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	-1.0	+7.2	+0.5	+6.8
2021	-0.4	-0.2	+1.6	-0.9	-0.2	+1.9	-0.1	+0.8	+0.1	+0.8	+3.2	-1.7	+4.8
2022	+0.1	+0.1	+2.6	-1.3	-0.3	-0.1	+1.9	-1.9	+0.1	+1.6	-0.9	+0.1	+2.3
2023	+1.5	+0.4	+1.3	+0.6	+2.9	+1.0	+1.0	+0.6	+0.3	+2.2	+3.4	+0.4	+15.6
2024	+2.8	+0.3	-0.4	+0.1	+0.9	+3.3	-1.4	+0.5	+0.9	+3.3	+0.9	-0.9	+10.3
2025	+0.5	+0.3	+0.1	+1.3	-1.1	+5.3	+0.2	+0.4	+0.7	+0.2	-0.5	+0.1	+7.5
2026	+0.3	+0.3	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.7

Variant C — Buy Call 4× + Money Market on remaining

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	-3.9	+28.8	+2.1	+27.0
2021	-1.5	-1.0	+6.2	-3.6	-0.8	+7.7	-0.5	+3.2	+0.3	+3.0	+13.0	-7.0	+19.0
2022	-0.0	+0.1	+10.1	-5.6	-1.7	-0.8	+7.1	-7.9	+0.1	+5.6	-4.0	+0.1	+3.2
2023	+5.0	+0.4	+4.0	+1.3	+10.6	+2.7	+2.7	+1.3	-0.0	+6.5	+12.5	+0.4	+47.3
2024	+10.2	-0.1	-2.9	-0.6	+2.5	+11.9	-6.6	+0.8	+1.1	+12.1	+2.3	-4.9	+25.9
2025	+0.8	+0.3	-0.4	+4.2	-5.3	+20.2	-0.2	+0.5	+0.8	-0.4	-2.9	-0.4	+17.1
2026	+0.3	+0.3	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.7

7. Key Take-Aways

- **Buy Call (Mode 2) is the structural winner** among the four trading modes: CAGR +5.0%, MaxDD –2.4%, Sharpe 1.00. The strategy profits from positive momentum while capping downside to the premium paid.
- **Sell Put (Mode 3) has lower CAGR (+2.9%) and larger drawdown (–8.0%)** driven by a concentrated loss period (Jul–Aug 2024). Despite a 65% win rate, the tail risk of short puts is evident.
- **Direction-neutral short premium (Mode 4) destroys value:** CAGR +0.7%, MaxDD –16.5%, Sharpe 0.10. Selling calls on down-momentum stocks adds directional risk in rallying markets.
- **Money Market allocation (Variant B) transforms the risk profile:** idle capital (~94% of notional) earns ~4–5% p.a., lifting CAGR from +5.0% to +7.3% and Sharpe from 1.00 to 1.55 with a *lower* MaxDD (–1.85%).
- **4× leverage (Variant C) delivers +17.1% CAGR** — the best absolute return — at a manageable MaxDD of –8.8%. Suitable for risk-tolerant allocators who can absorb short-term volatility (monthly std dev ~6.2%).
- **Dynamic IV-rank filtering** (Top 30 from SPX 50) concentrates exposure in the highest-premium names, improving expected premium collection relative to a static universe.
- **Operational fit:** the strategy is fully systematic, option-expiry driven (bi-monthly rebalance), and requires no intraday monitoring. Lambda-based automation handles signal generation and S3 storage.

Disclaimer: This document is for internal research purposes only. Past backtest performance is not indicative of future results. Options trading involves significant risk of loss.